

AMENDMENT NO. 1 SEPTEMBER 2022
TO
IS 17525 : 2021 SPECIFICATION FOR BULLET
RESISTANT MORCHA

[Page 2, Table 1, Sl No. ix), col 3, b] — Substitute ‘1 350 ± 5 mm and 750 ± 5 mm’ for ‘1 350 mm and 750 mm’

[Page 2, Table 1, Sl No. viii), col 3] — Substitute the following for existing:

‘All hinge joints (including pin and chain) of bullet resistant morcha shall have min of 5 mm thickness and length and breadth as per user requirement.’

[Page 2, Table 1, Sl No. ix), col 3, c] — Substitute ‘750 ± 5 mm’ for ‘750 mm’

(Page 6, clause 7.4.1) — Substitute the following for existing:

‘7.4.1 Fair Hit Criteria

Unless specified otherwise, a shot is considered fair hit, if it impacts the panel and meets the following criteria:

7.4.1.1 Impact velocity

- a) Impact velocity is within the limits specified in Table 5.
- b) Impact velocity is higher than the limit specified in Table 5, but perforation is not observed in the witness plate as per 7.4.3.
- c) Impact velocity is lower than the limit specified in Table 5, but perforation is observed in the witness plate as per 7.4.3.

7.4.1.2 Edge to shot distance

- a) Edge to shot distance is within the limits specified in Table 5 and Table 6.
- b) Edge to shot distance is lower than the limit specified in Table 5 and Table 6, but perforation is not observed in the witness plate as per 7.4.3.

7.4.1.3 Inter shot distance

- a) Inter shot distance is within the limits specified in Table 5 and Table 6.
- b) Inter shot distance is lower than the limit specified in Table 5 and Table 6, but perforation is not observed in the witness plate as per 7.4.3.

In case of invalid shots, additional shots may be fired on a test sample depending upon the area available and possibility of fair hit. If the sufficient area is not available for additional shots, the sample shall be discarded and a fresh sample shall be taken for ballistic testing.’

(Page 7, clause 7.4.3) — Substitute the following for existing:

‘a) A witness plate of aluminium alloy AA 2024 T3 (or T4) with thickness 0.5 ± 0.05 mm shall be placed at a stand off distance of 150 ± 10 mm behind and parallel to the back face surface of the target at the aim point. The witness system should extend over a sufficient area (equal to or larger than the target size) such that all significant projectile or target debris can be detected.

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The level of damage on the aluminium witness plate shall constitute the acceptance criterion for the evaluation of success or failure of a target against a particular projectile. After each ballistic impact, the witness plate shall be visually examined for damage and evidence of penetration by the projectile or target materials. Perforation of the target shall be recorded when light is observed to pass through the damage in the witness plate and the sample shall be rejected.

b) After each ballistic test, the rear surface of the specimen shall be examined for the presence of any crack/tear. Perforation of the target shall be recorded if any crack/tear is observed permitting the passage of kerosene oil.'

(Page 7, Table 7, col 3, Sl No. 3) — Add the following:

‘(4 shots on each panel)’

(Page 7, clause 7.5.3, last line) — Substitute the following for existing:

‘The requirement of minimum edge-to-shot distance is not applicable for structurally weak areas.

NOTE — User may decide the impact location to exploit the potential weak area of the target.’

(Page 8, clause 7.5.4, last line) — Substitute the following for existing:

‘Acceptance criteria will be same as prescribed in 7.4.3 a).’

(Page 8, clause 7.5.6) — Substitute the following for existing:

‘Acceptance criteria will be same as prescribed in 7.4.3 a).’

(Page 8, clause 7.5.5) — Substitute the following for existing:

‘Unless specified otherwise, a shot is considered fair hit, if it impacts the panel and meets the following criteria:

7.5.5.1 Impact velocity

- a) Impact velocity is within the specified limits.
- b) Impact velocity is higher than the limit specified, but perforation is not observed in the witness plate.
- c) Impact velocity is lower than the specified limit, but perforation is observed in the witness plate.

7.5.5.2 Edge to shot distance

- a) Edge to shot distance is within the specified limits.
- b) Edge to shot distance is lower than the specified limits, but perforation is not observed in the witness plate as per 7.4.3.

7.5.5.3 Inter shot distance

- a) Inter shot distance is within the specified limits.
- b) Inter shot distance is lower than the specified limits, but perforation is not observed in the witness plate as per 7.4.3.

In case of invalid shots, additional shots may be fired on a test sample depending upon the area available and possibility of fair hit. If the sufficient area is not available for additional shots, the sample shall be discarded and afresh sample shall be taken for ballistic testing.’

(Page 10, clause 8.2) — Substitute the following for existing:

‘Unless specified otherwise, a shot is considered fair hit, if it impacts the panel and meets the following criteria:

8.2.1 Impact Velocity

- a) Impact velocity is within the limits specified in Table 4.
- b) Impact velocity is higher than the limits specified in Table 4, but perforation is not observed in the witness plate.

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- c) Impact velocity is lower than the limits specified in Table 4, but perforation is observed in the witness plate.

8.2.2 Edge to Shot Distance

- a) Edge to shot distance is within the limits specified in Table 12.
b) Edge to shot distance is lower than the limits specified in Table 12, but perforation is not observed in the witness plate as per **7.4.3**.

8.2.3 Inter Shot Distance

- a) Inter shot distance is within the limits specified in Table 12.
b) Inter shot distance is lower than the limits specified in Table 12, but perforation is not observed in the witness plate as per **7.4.3**.
c) Inter shot distance is higher than the limits specified in Table 12, but perforation is observed in the witness plate as per **7.4.3**.

In case of invalid shots, additional shots may be fired on a test sample depending upon the area available and possibility of fair hit. If the sufficient area is not available for additional shots, the sample shall be discarded and afresh sample shall be taken for ballistic testing.'

(Page 10, clause **8.3**) — Substitute the following for existing:

'Acceptance criteria will be same as prescribed in **7.4.3 a)**.'

(Page 10, clause **9.1.2**, last line) — Add the following:

'The user may ask for one complete sample of fully fabricated morcha for evaluation during the tender itself. This Morcha shall be subjected to ballistic testing as per **7.5**. No major/critical defects mentioned in **9.4** shall be acceptable.'

(Page 10, clause **9.4.1**) — Substitute the following for existing:

'Perforation of witness plate as specified in **7.4.3 a)**.'